

K hyperparameter, FINMEM can expand this capacity far beyond the human cognitive scope. We aimed to determine whether such flexibility in adjusting memory bandwidth translates to improvements in FINMEM’s performance.

Metric	B&H	Top 1	Top 3	Top 5	Top 10
Cumulative Return (%)	-66.9497	52.0936	29.4430	54.6958	79.4448
Sharpe Ratio	-2.0845	1.8642	1.1214	2.4960	2.7469
Daily Volatility (%)	3.8050	3.3105	3.1105	2.5960	3.4262
Annualized Volatility (%)	60.4020	52.5529	49.3779	41.2100	54.3891
Max Drawdown (%)	67.3269	25.2355	27.0972	12.5734	17.1360

Table 5: Comparison of overall trading performance during the testing period with different configurations of working memory capacity.

As demonstrated in Table 5, we adjusted the hyperparameter K to alter the number of memory events retrieved from shallow, intermediate, and deep long-term memory layers in FINMEM. We tested K values of 1, 3, 5, and 10, exploring FINMEM’s working memory capabilities at levels below, near, and above the human cognitive limit. For all these K settings, we maintained a self-adaptive risk inclination, while other settings were consistent with those described in Section 5.1.

Across all K configurations, FINMEM outperformed the Buy & Hold baseline, indicating the effectiveness of its memory module in processing diverse information and capturing critical events, which subsequently enhanced its trading performance, as evidenced by positive Cumulative Returns and Sharpe Ratios. Notably, higher K values, like 5 and 10, enabled FINMEM to achieve the best Cumulative Returns and Sharpe Ratios exceeding 2.0. With K set to 1, FINMEM still performed moderately well by capturing the most critical memory events of each layer.

An in-depth analysis in Figure 11, which shows the Cumulative Return over time for various K settings, reveals that a K value of 5 is optimal for trading TSLA stock, consistently delivering robust performance with the lowest Volatility and Max-Drawdown. Before mid-October 2022, when the stock market was relatively stable and slightly upward, FINMEM’s trading actions aligned well with market trends (referring to B&H) and avoided significant losses. During periods of high volatility and continuous downturns (post-mid-October 2022), it maintained earnings by reducing “Buy” actions and favoring more “Hold” and “Sell” strategies. However, setting K to 10, while effective during market volatility, resulted in significant losses in stable market conditions. The issue may stem from the disproportionately loose capacity constraints on incoming information relative to the volume of incoming data. The broad memory retrieval bandwidth might have mixed trivial messages with critical ones, hampering FINMEM’s decision precision. This challenge becomes especially evident in neutral market conditions, where the influx of information includes a mix of varying market sentiments and trends.

Addressing **RQ5**, appropriately tuning the number of memory events (Top- K in the FINMEM memory module can significantly enhance its trading performance. The aforementioned study illustrates that FINMEM can achieve optimal results by effectively assimilating key signals from a sufficient quantity of filtered memories across each layer. However, the optimal value for K may vary depending on the volume and quality of incoming information.

7 Conclusion and future work

In this paper, we introduce FINMEM, an innovative automated trading agent framework featuring an adjustable cognitive memory structure and dynamic character design. Our research demonstrates its capacity to enhance stock trading performance substantially using real-world financial datasets. Additionally, the efficacy of each critical component within FINMEM is thoroughly demonstrated in our ablation studies, highlighting their roles in optimizing trading outcomes.

Its unique features of human-like cognitive memory modules and dynamic character design enable it to tackle the complexities of financial environments and respond aptly to new situations. Compared to other LLM trading agents, FINMEM’s memory module equips it with the capability to better comprehend financial data featured by various timeliness and organize them as a self-evolving long-term memory layer. The dynamic character design endows FINMEM with critical professional insights, enabling efficient filtering of impactful messages from incoming financial

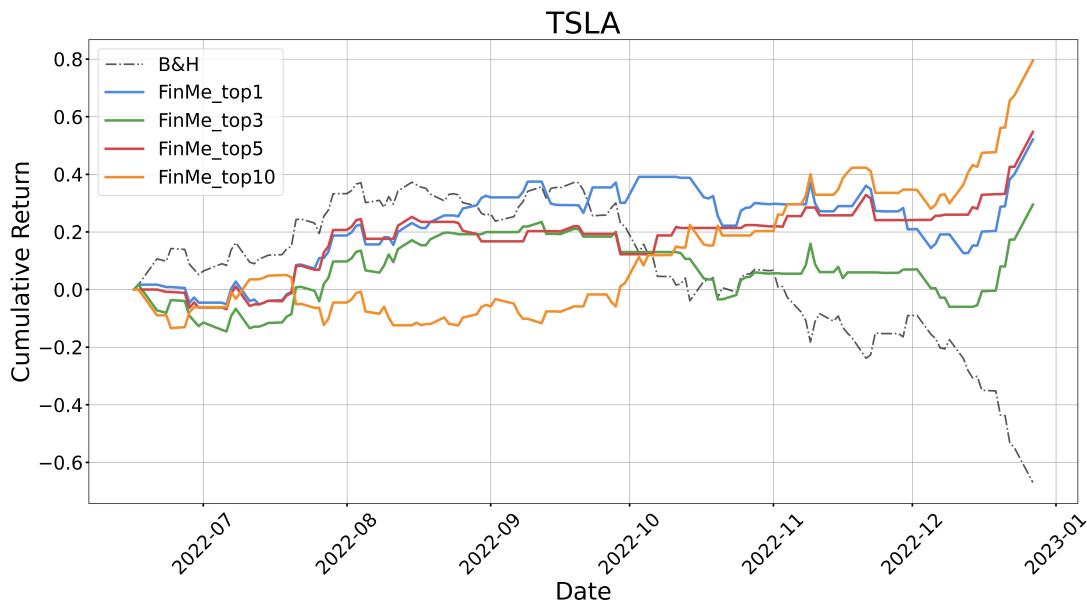


Figure 11: Cumulative Return over time for with different FINMEM working memory capacity.

data for trading actions. Additionally, the integration of multiple risk profiles enhances FINMEM’s adaptability to a range of market conditions.

FINMEM’s exceptional performance underscores its remarkable ability to transform diverse financial data into well-informed investment strategies. Its proficiency in integrating various financial data types is further accentuated by a notably reduced training duration, which offers advantages for trading with newly established companies. In our approach, we utilized a limited range and quality of financial news and reports, employing general-purpose LLMs as the backbone algorithms. However, FINMEM is fully compatible with LLMs specifically fine-tuned for financial applications. We anticipate that its trading efficacy will be elevated further with access to a more comprehensive and higher-quality dataset, coupled with LLMs tailored specifically for financial contexts.

While primarily designed for financial decision-making, the FINMEM framework boasts a versatility that extends to domains such as IT consulting and business reporting, where actions are driven by time-sensitive information. Looking ahead, an intriguing direction for development is the creation of a multi-agent trading system, rooted in the FINMEM platform, aimed at enhancing investment portfolio optimization. This system, featuring diverse professional backgrounds in its profiling modules, enables concurrent operations and trading across a variety of financial products. It dynamically adjust trading targets based on a sequential assessment of key trading performance indicators. Simultaneously, it can timely reallocate investment proportions for each financial product category, leveraging peer-to-peer communication and systematic performance analysis. Supported by LLMs, this multi-agent trading system is poised to assemble an optimized investment portfolio through its strategically planned operations.

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